

Omer Sabri Emeksiz

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Education

Koç University

Ph.D. in Computer Science and Engineering (Upcoming)
Advisor: Prof. Yücel Yemez

Istanbul, Turkey

September 2026

Research Focus: Computer Vision

Koç University

M.Sc. in Computer Science and Engineering, GPA: 3.92/4.00
Advisor: Prof. Metin Sezgin, Prof. Yücel Yemez, Prof. Engin Erzin

Istanbul, Turkey

October 2024 – June 2026 (Expected)

Research Focus: EEG-based cybersickness in immersive VR; Deep RL for affective computing on social robots

Marmara University

B.Sc. in Electrical and Electronic Engineering, GPA: 3.33/4.00 (Honor, Ranked 5th)
Advisor: Prof. Engin Maşazade

Istanbul, Turkey

September 2021 – June 2024

Research Focus: Reinforcement Learning

Publications

- Ö. S. Emeksiz, Y. Yemez, E. Erzin, T. M. Sezgin, “[Early Prediction of Cybersickness from EEG in Immersive VR](#)”, IEEE International Conference on Multimedia and Expo (ICME), 2026 (accepted)
- Ö. S. Emeksiz, E. Erzin, Y. Yemez, T. M. Sezgin, “[A Convolutional Transformer Model for EEG-Driven Cybersickness Detection in VR Experience](#)”, IEEE 33rd Signal Processing and Communications Applications Conference (SIU), 2025
- Ö. S. Emeksiz, E. Maşazade, S. Selim, “[AI-Driven Optimization of the Filling Process: A Comparison of Reinforcement Learning Methods](#)”, IEEE 33rd Signal Processing and Communications Applications Conference (SIU), 2025
- Ö. S. Emeksiz, M. E. Ağcabay, E. Maşazade, S. Selim, T. Boysan, “[A Reinforcement Learning Model for Industrial Filling Process Control](#)”, IEEE 30th International Conference on Electronics, Circuits and Systems (ICECS), 2023

Research Experience

KUIS AI Center, Koç University

AI Researcher, IUI Lab & MVGL Lab

Istanbul, Turkey

October 2024 – Present

- Conducting research on EEG-based cybersickness prediction in immersive VR using time series modeling and computer vision. **Published at ICME 2026, SIU 2025.**
- Leading a multimodal EEG data collection study to establish an open benchmark for cybersickness research, addressing the lack of publicly available datasets in the field.
- Research supported by [European CHIST-ERA GENESIS](#) (CHIST-ERA-20-BCI-003) and TÜBİTAK under Project No. 221N367.
- Conducting research on deep reinforcement learning and affective computing for autonomous facial expression discovery on social robots. **Under review at a top ML venue.**
- Designed and implemented **Wrap-Up!**, an AI-powered collaborative meeting visualization tool: a full-stack Chrome extension for Google Meet that transcribes live audio (STT), orchestrates an LLM to extract meeting structure, and renders a shared real-time canvas with proactive AI facilitation and thread tracking.
- Supervised by [Prof. Metin Sezgin](#), [Prof. Yücel Yemez](#), and [Prof. Engin Erzin](#).

Marmara University

Undergraduate AI Researcher

Istanbul, Turkey

December 2022 – June 2024

- Conducted research on reinforcement learning for industrial filling process control, formulating the MDP framework and optimizing feed switching policy using Monte Carlo methods. **Published at ICECS 2023, SIU 2025.**
- Supervised by [Prof. Engin Maşazade](#).

Research Interests

Reinforcement Learning · Time Series Modeling · Computer Vision · Affective Computing

Work Experience

Baykon Industrial Weighing Systems

AI Engineer

Istanbul, Turkey

February 2024 – October 2024

- Developed adaptive RL control policies (Monte Carlo, TD, Q-learning, MAB) achieving optimal feed switching for industrial filling process.
- Collected an industrial filling dataset (1,000 filling runs) and deployed the system on an embedded filling machine.
- Led and completed the [TUBITAK-1501](#) industrial AI project.

TUBITAK

Engineer Intern

Istanbul, Turkey

August 2023 – September 2023

- Developed a continuous-time TCP/IP communication application using C# and a data extraction tool using Python.

Open Business Software Solutions (OBSS)

Software Engineer Intern

Istanbul, Turkey

July 2023 – August 2023

- Developed a full-stack enterprise-level [Human Resources Job Postings and Applications Management System](#) web application using Java, Spring Boot for backend and ReactJS, MUI for frontend.
- Used LDAP and LinkedIn API with Spring Security and JWT for authentication and authorization.
- Managed project with Apache Maven and utilized Hibernate, Spring Data JPA and MySQL for database.

Kalem Software

Software Engineer Intern

Istanbul, Turkey

August 2022 – September 2022

- Developed backend services for two B2B client web applications: [File Organizer](#), a file storage and management platform, and [Material Waybill Organizer](#), a system for materials organization, using Java and Spring Boot.
- Managed project with Apache Maven and utilized Hibernate, Spring Data JPA and MySQL for database.

Professional Service

- **Reviewer:** IEEE ICME 2026, IEEE TMM, IEEE SIU 2026
- **Teaching Assistant:** Introduction to Computer Science and Programming; Computer Engineering Design; Communication Engineering

Relevant Coursework

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| • Computer Vision | • Intelligent User Interfaces | • Biomedical Signal Processing |
| • Deep Learning | • Medical Image Analysis | • AI in Health Sciences |
| • Deep Unsupervised Learning | • Embodied Intelligence | • Image Processing |

Skills

- **Programming Languages:** Python, C, C++, Java
- **Libraries and Frameworks:** PyTorch, NumPy, Scikit-learn, OpenCV, HuggingFace, Spring Boot, ReactJS